REPUBLIC POLYTECHNIC

C339 Software Testing and Analytics
Continuous Assessment 1 (CA1)

Project 03: Parking Meter System Requirements Specification

1. About

A car park requires a simple software system to calculate parking charges and check whether a short stay permit is valid.

The system consists of two independent functions:

- **Calculate Parking Charge:** Calculate the charge for a stay measured in whole minutes.
- **Determine Permit Validity:** Determine whether an active permit covers the parking stay.

Both functions must operate according to the requirements described below.

2 General Requirements

The following requirements apply to both functions:

1. All monetary amounts, where used, are represented in Singapore cents (SGD), using whole numbers without decimal places.
2. Numeric inputs must be Python integers (int). Boolean values are not valid integers. A parameter explicitly declared as bool accepts only True or False.
3. If an argument has an invalid data type, the function must raise a **TypeError**.
4. If an argument has the correct data type but falls outside the permitted range, the function must raise a **ValueError**.
5. The functions operate independently and do not require a database, network connection, or shared application state.

3 Functional Requirements

3 1 Calculate Parking Charge

Source File: parking_fee.py

Function: calculate_parking_fee(minutes)

Description

Calculate the charge for a stay measured in whole minutes.

Parameter	Data Type	Constraints
minutes	int	Must be a nonnegative integer

Requirements

1. Charge 200 cents for each started hour. A stay of 1 to 60 minutes costs 200 cents; 61 to 120 minutes costs 400 cents, and so on.
2. Zero minutes costs zero. Any part of an hour must be charged as a full hour.

Output: Return the parking charge as an integer in cents.

3 2 Determine Permit Validity

Source File: parking_permit.py

Function: is_permit_valid(permit_active, parked_minutes)

Description

Determine whether an active permit covers the parking stay.

Parameter	Data Type	Constraints
permit_active	bool	Must be a Python bool
parked_minutes	int	Must be a nonnegative integer

Requirements

1. The permit is valid only when permit_active is True and parked_minutes is at most 120, including zero.
2. An inactive permit or a stay above 120 minutes is not valid.

Output: Return the actual True or False value. The integers 1 and 0 are not acceptable return values.